

Design and Implementation of a Blockchain-Based E-Voting System

TEAM MEMBERS :

S.THILLAIARASAN (620822205114)

M.RAMANA (620822205083)

V.SUGUMARAN (620822205107)

K.UDHAY (620822205117)

GUIDE : MR. JEEVA. D, AP/IT

Problem Statement

- ▶ Existing electronic voting systems lack transparency and verifiability for voters.
- ▶ Voters cannot independently verify whether their votes are recorded correctly.
- ▶ Centralized databases are vulnerable to cyber-attacks and unauthorized modifications.
- ▶ Manual verification and audit processes are time-consuming and error-prone.
- ▶ There is a need for a secure, transparent, and verifiable voting system that minimizes human intervention.
- ▶ The challenge is to design a system that ensures one-person-one-vote while preserving data integrity.

Objective

- ▶ To design a secure and transparent electronic voting system using blockchain technology.
- ▶ To prevent vote tampering and duplicate voting using smart contracts.
- ▶ To enable verifiable and immutable vote storage.
- ▶ To reduce reliance on centralized election authorities.
- ▶ To provide a simple and user-friendly interface for voters.
- ▶ To demonstrate a real-world application of blockchain in governance systems.

Existing System

- ▶ Centralized electronic voting infrastructure.
- ▶ Election data stored in traditional databases.
- ▶ Vote counting and result generation controlled by authorities.
- ▶ Limited transparency for voters.
- ▶ Manual audits required to verify results.

Disadvantage

- ▶ Single point of failure.
- ▶ Susceptible to hacking and data manipulation.
- ▶ Lack of voter trust and transparency.
- ▶ Difficult to audit independently.
- ▶ High operational and verification costs.

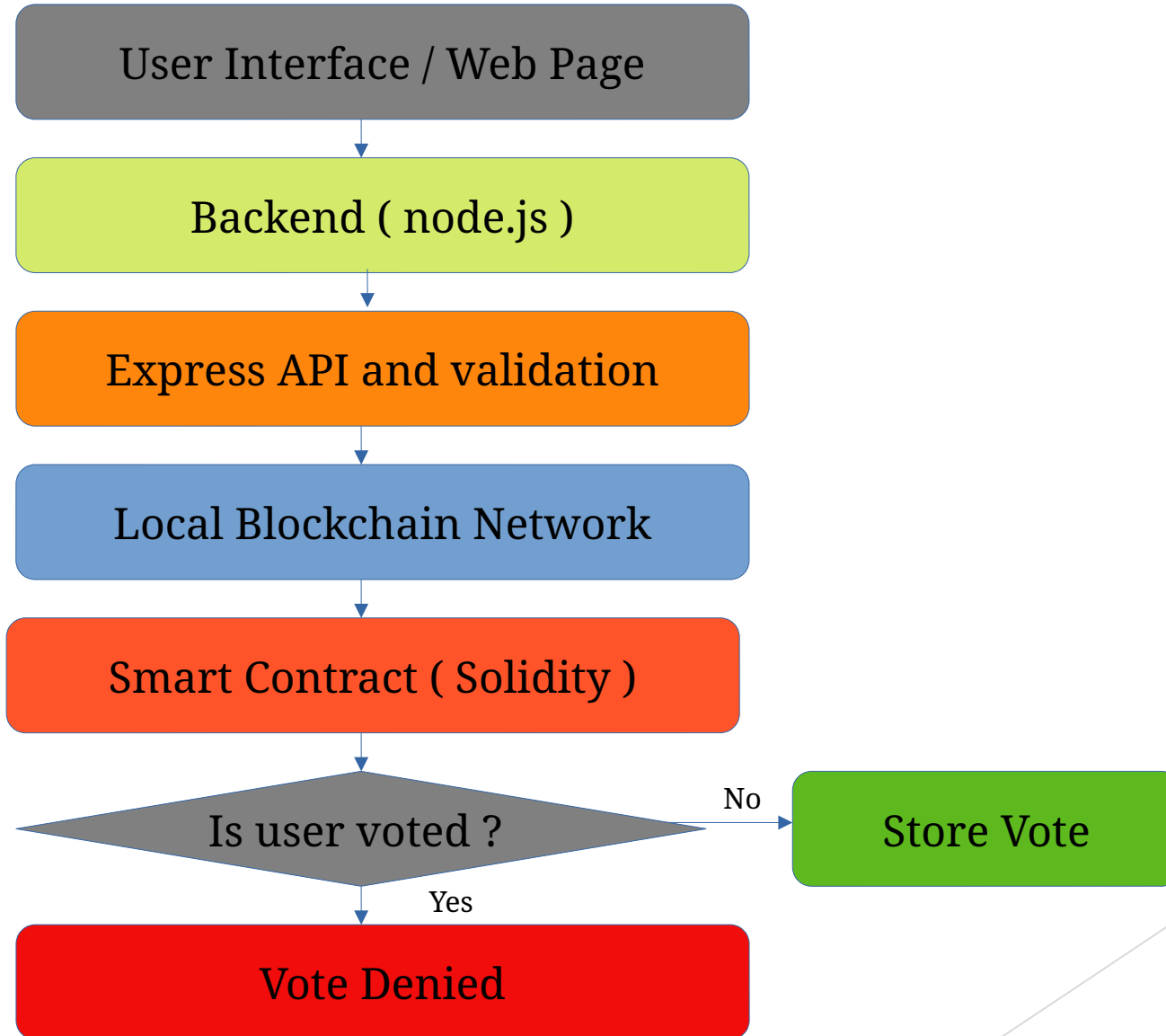
Proposed System :

- ▶ Blockchain-based decentralized voting system.
- ▶ Smart contracts manage voter registration and voting logic.
- ▶ Votes stored on blockchain ensuring immutability.
- ▶ Transparent result computation.
- ▶ Automated enforcement of election rules.

Advantage

- ▶ No single point of failure.
- ▶ Tamper-proof and immutable vote records.
- ▶ Improved transparency and trust.
- ▶ Reduced manual verification.
- ▶ High security through cryptographic techniques.

SYSTEM ARCHITECTURE/FLOW DIAGRAM



Modules :

- ▶ User Interface Module
- ▶ Voter Registration Module
- ▶ Voting Module
- ▶ Vote Verification Module
- ▶ Result Management Module

MODULE 1: User Interface Module

- ▶ Purpose: Provide a secure, intuitive front-end for all users (voters, admins, auditors).
- ▶ Key Responsibilities:
 - Present registration, voting, and result views clearly.
 - Validate input before sending data to backend / blockchain.
 - Handle authentication flows and user feedback (errors, confirmations).
- ▶ Key Features:
 - Responsive design for desktop and mobile.
 - Multi-language support (if in scope).
 - Accessibility-compliant layouts and controls.
- ▶ Technology Stack :
 - Web: Javascript, Nodejs
 - Styling: CSS / TailwindCSS .
 - Integration: Express, Ganache.

MODULE 2: Voter Registration Module

- ▶ Purpose: Onboard eligible voters and ensure only valid identities can cast votes.
- ▶ Key Responsibilities:
 - Capture voter identity and eligibility details securely.
 - Interface with external ID/registry systems (if applicable).
 - Issue unique voter credentials / keys linked to blockchain identity.
- ▶ Core Flows:
 - New voter registration and verification.
 - Updating voter details (within configured rules).
 - De-duplication and fraud detection checks.
- ▶ Security & Compliance:
 - KYC / legal requirements (jurisdiction-specific).
 - Encryption of PII, strict access controls, and audit logs

MODULE 3: Voting Module

- ▶ Purpose: Enable voters to cast ballots securely and anonymously via the blockchain.
- ▶ Key Responsibilities:
 - Present live ballots and options to authenticated, eligible voters.
 - Validate voter eligibility and “one person, one vote” rules.
 - Submit votes as blockchain transactions with end-to-end integrity.
- ▶ Core Flows:
 - Ballot selection and confirmation (review screen).
 - Vote encryption and signing with voter keys.
 - Handling network / transaction failures and retries.
- ▶ Quality & Security:
 - Strong cryptography for ballot secrecy.
 - Protection against double-voting, replay, and tampering.

Conclusion

- ▶ The proposed blockchain-based voting system addresses major drawbacks of traditional electronic voting.
- ▶ Decentralization eliminates single points of failure and increases system reliability.
- ▶ Smart contracts enforce election rules automatically without human intervention.
- ▶ The system enhances transparency, security, and voter trust.
- ▶ The prototype demonstrates the practical feasibility of blockchain in electronic voting.
- ▶ This work lays the foundation for future large-scale and national-level implementations.



THANK YOU !